

Jenkins Pipelines Reference

All Jenkinsfiles from `jenkins-tutorial-main`, consolidated into one file for reference. Each section below is a **separate, independent** Jenkinsfile — Jenkins only supports one `pipeline {}` / `node {}` block per file, so these can't be merged into a single running pipeline. Copy the block you need into a file named `Jenkinsfile` in the relevant project directory.

Two files had broken brace nesting in the original upload (`7-Environment-Variables/Jenkinsfilev2` and `5-Agent-Setup/Jenkinsfile`) — those are fixed below, and the fix is noted inline.

1. Declarative pipeline — basic 3-stage

`2-Jenkins-pipeline/declarative-pipeline/jenkinsfile-declarative`

```
pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                sh 'echo "Building project..."'
            }
        }
        stage('Test') {
            steps {
                sh 'echo "Running tests..."'
            }
        }
        stage('Deploy') {
            steps {
                sh 'echo "Deploying application..."'
            }
        }
    }
}
```

2. Declarative pipeline — hello world

2-Jenkins-pipeline/declarative-pipeline/jenkinsfile-declarative-ws

```
pipeline {
    agent any

    stages {
        stage('Hello') {
            steps {
                sh 'echo "Hello World"'
            }
        }
    }
}
```

3. Scripted pipeline — basic 3-stage

2-Jenkins-pipeline/scripted-pipeline/jenkinsfile-scripted

```
node {
    stage('Build') {
        sh 'echo "Building project..."'
    }
    stage('Test') {
        sh 'echo "Running tests..."'
    }
    stage('Deploy') {
        sh 'echo "Deploying application..."'
    }
}
```

4. Scripted pipeline — hello world

2-Jenkins-pipeline/scripted-pipeline/jenkinsfile-scripted-ws

```
node {
    stage('Hello') {
        sh 'echo "Hello World"'
    }
}
```

```
}
```

5. Environment variables — credentials binding demo

7-Environment-Variables/Jenkinsfile

```
pipeline {
    agent any

    stages {
        stage('Hello') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'DockerHub',
                    passwordVariable: 'DockerPass',
                    usernameVariable: 'DockerUser'
                )]) {
                    echo "Hello World"
                    echo "Docker username: ${DockerUser}"
                    // sh "docker login -u ${DockerUser} -p ${DockerPass}"
                }
            }
        }
    }
}
```

6. Environment variables — Docker pull/tag/push

7-Environment-Variables/Jenkinsfilev2 *(fixed: original had one unbalanced closing brace after the Docker Push stage, which would fail to parse)*

```
pipeline {
    agent any

    environment {
        APP_NAME      = 'theshubhamgour/python-webapp'
        TAG_VERSION   = 'v1.0.0'
        IMAGE          = 'theshubhamgour/python-app:v1'
    }

    stages {
```

```

stage('Docker Image Pull') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'DockerHub',
            passwordVariable: 'DockerPass',
            usernameVariable: 'DockerUser'
        )]) {
            echo "Pulling Docker image using credentials"
            sh "docker login -u ${DockerUser} -p ${DockerPass}"
            sh "docker pull ${IMAGE}"
            echo "Docker Pull completed successfully"
        }
    }
}

stage('Docker Image Tag') {
    steps {
        echo "Docker image tagging in progress"
        sh "docker tag ${IMAGE} ${APP_NAME}:${TAG_VERSION}"
        echo "Docker tag completed successfully"
    }
}

stage('Docker Push') {
    steps {
        echo "Docker image Push in progress"
        sh "docker push ${APP_NAME}:${TAG_VERSION}"
        echo "Docker Push successfully"
    }
}
}

```

7. Java + Maven + Docker

3-java-jenkins-docker-app/Jenkinsfile

```

pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }
    }
}

```

```

    stage('Test') {
        steps {
            sh 'mvn test'
        }
    }
    stage('Docker Build') {
        steps {
            sh 'docker build -t java-jenkins-docker:latest .'
        }
    }
    stage('Docker Run') {
        steps {
            sh 'docker run --rm java-jenkins-docker:latest'
        }
    }
}
}

```

8. Python + Docker — clone, build, test, push

4-python-jenkins-docker-app/Jenkinsfile

```

pipeline {
    agent any

    stages {
        stage('Clone Github Repository') {
            steps {
                git branch: 'main', changelog: false, poll: false,
                    url: 'https://github.com/theshubhamgour/jenkins-tutorial.git'
                sh 'pwd'
            }
        }
        stage('Build') {
            steps {
                dir('4-python-jenkins-docker-app') {
                    sh 'python3 -m py_compile app.py'
                }
            }
        }
        stage('Test') {
            steps {
                dir('4-python-jenkins-docker-app') {
                    sh 'python3 -m unittest discover tests'
                }
            }
        }
    }
}

```

```

    }
}
stage('Docker Build') {
    steps {
        dir('4-python-jenkins-docker-app') {
            sh 'docker build -t python-app:latest .'
        }
    }
}
stage('Docker Run') {
    steps {
        dir('4-python-jenkins-docker-app') {
            sh 'docker run --rm python-app:latest'
        }
    }
}
stage('Docker Tag & Push') {
    steps {
        dir('4-python-jenkins-docker-app') {
            withCredentials([usernamePassword(
                credentialsId: 'theshubhamgour',
                passwordVariable: 'DockerhubPassword',
                usernameVariable: 'DockerhubUserName'
            )]) {
                sh '''
                    echo "$DockerhubPassword" | docker login -u "$Dockerh
                    docker tag python-app:latest theshubhamgour/python-ap
                    docker push theshubhamgour/python-app:latest
                '''
            }
        }
    }
}

post {
    always {
        cleanWs()
    }
    success {
        echo 'Pipeline completed successfully!'
    }
    failure {
        echo 'Pipeline failed!'
    }
}

```

```
}
```

9. Agent setup — Java build on a worker node

5-Agent-Setup/Jenkinsfile *(fixed: original had mismatched closing braces around the Docker Push stage; also `docker login -p {password}` was a literal placeholder, replaced here with a credentials binding)*

```
pipeline {
    agent { label 'worker-node' }

    stages {
        stage('Build') {
            steps {
                git branch: 'main', changelog: false, poll: false,
                    url: 'https://github.com/theshubhamgour/jenkins-tutorial.git'
                dir('3-java-jenkins-docker-app') {
                    sh "mvn clean package"
                }
            }
        }
        stage('Test') {
            steps {
                dir('3-java-jenkins-docker-app') {
                    sh "mvn test"
                }
            }
        }
        stage('Compile') {
            steps {
                dir('3-java-jenkins-docker-app') {
                    sh "mvn compile"
                }
            }
        }
        stage('Docker Build') {
            steps {
                dir('3-java-jenkins-docker-app') {
                    sh "docker build -t java-app ."
                }
            }
        }
        stage('Docker Run') {
            steps {
```

```

        dir('3-java-jenkins-docker-app') {
            sh "docker run java-app"
        }
    }
}
stage('Docker Push') {
    steps {
        dir('3-java-jenkins-docker-app') {
            withCredentials([usernamePassword(
                credentialsId: 'theshubhamgour',
                passwordVariable: 'DockerPass',
                usernameVariable: 'DockerUser'
            )]) {
                sh "docker login -u ${DockerUser} -p ${DockerPass}"
                sh "docker tag java-app theshubhamgour/java-app:latest"
                sh "docker push theshubhamgour/java-app:latest"
            }
        }
    }
}
}
}
}

```

10. Agent execution — Python build on a worker node

6-Agent-execution/python-app/Jenkinsfile

```

pipeline {
    agent { label 'worker-node' }

    stages {
        stage('Clone Github Repository') {
            steps {
                git branch: 'main', changelog: false, poll: false,
                    url: 'https://github.com/theshubhamgour/jenkins-tutorial.git'
                sh 'pwd'
            }
        }
        stage('Build') {
            steps {
                dir('4-python-jenkins-docker-app') {
                    sh 'python3 -m py_compile app.py'
                }
            }
        }
    }
}

```

```

    }
    stage('Test') {
        steps {
            dir('4-python-jenkins-docker-app') {
                sh 'python3 -m unittest discover tests'
            }
        }
    }
    stage('Docker Build') {
        steps {
            dir('4-python-jenkins-docker-app') {
                sh 'docker build -t python-app:latest .'
            }
        }
    }
    stage('Docker Run') {
        steps {
            dir('4-python-jenkins-docker-app') {
                sh 'docker run --rm python-app:latest'
            }
        }
    }
    stage('Docker Tag & Push') {
        steps {
            dir('4-python-jenkins-docker-app') {
                withCredentials([usernamePassword(
                    credentialsId: 'theshubhamgour',
                    passwordVariable: 'DockerhubPassword',
                    usernameVariable: 'DockerhubUserName'
                )]) {
                    sh '''
                        echo "$DockerhubPassword" | docker login -u "$Dockerh
                        docker tag python-app:latest theshubhamgour/python-ap
                        docker push theshubhamgour/python-app:latest
                    '''
                }
            }
        }
    }
}

post {
    always {
        cleanWs()
    }
    success {
        echo 'Pipeline completed successfully!'
    }
}

```

```
    }  
    failure {  
        echo 'Pipeline failed!'  
    }  
}  
}
```